Cancer patient survival can be accurately parameterized, improving trial precision and revealing time-dependent therapeutic effects

Deborah Plana[1,2], Geoffrey Fell[3], Brian M. Alexander[3,4], Adam C. Palmer[5*], Peter K. Sorger[1*]

[1]Laboratory of Systems Pharmacology and the Department of Systems Biology, Harvard Medical School, Boston, Massachusetts, USA. [2]Harvard-MIT Division of Health Sciences and Technology, Harvard Medical School and MIT, Cambridge, Massachusetts, USA.[3]Dana-Farber Cancer Institute, Boston, Massachusetts, USA. [4]Foundation Medicine Inc., Cambridge, Massachusetts, USA. [5]Department of Pharmacology, Computational Medicine Program, Lineberger Comprehensive Cancer Center, University of North Carolina at Chapel Hill, Chapel Hill, North Carolina, USA. *These authors contributed equally

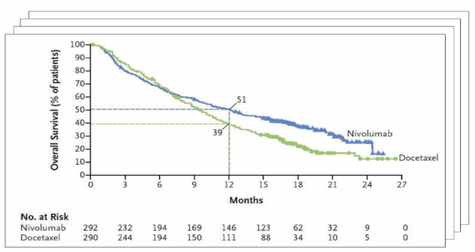
Laboratory of Systems Pharmacology

Key Points

- Individual participant data (IPD) from completed oncology clinical trials are rarely available.
- We imputed survival IPD from ~500 arms of phase III oncology trials from 2014-2016 in breast, colorectal, lung, and prostate cancer (representing ~220,000 events) and found that they are well fit by a two-parameter Weibull distribution.
- Weibull fitting increases the precision of drug efficacy estimates obtained from small patient cohorts typical of phase 1/2 trials.
- Trials often violate the proportional hazards assumption, especially those testing immune-checkpoint inhibitors, indicating time-dependent therapeutic effects.
- Clinical trial length has an underappreciated impact on the likelihood of trial success.

Methods

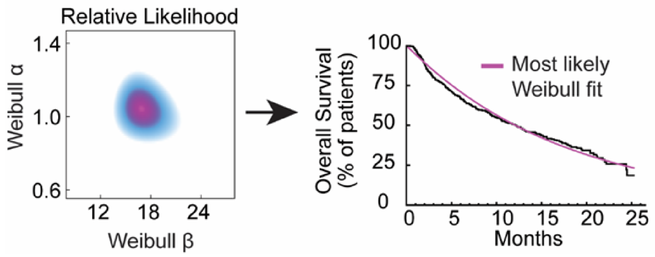
1) Plots from trial publications were used to impute IPD.



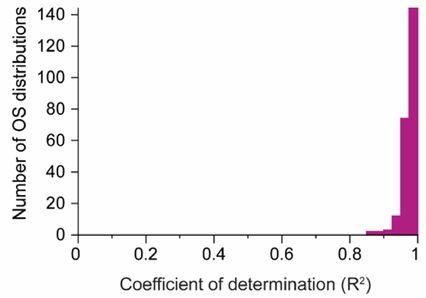
Nivolumab:

Event time (t, months)	Event type
5.1	Death
14.3	Death
...	...
7.5	Censoring
12.7	Censoring
... for 292 events	

2) The likelihood of a distribution to describe IPD was computed for a set of parameter values. The parameters with the highest likelihood best describe trial arm IPD.

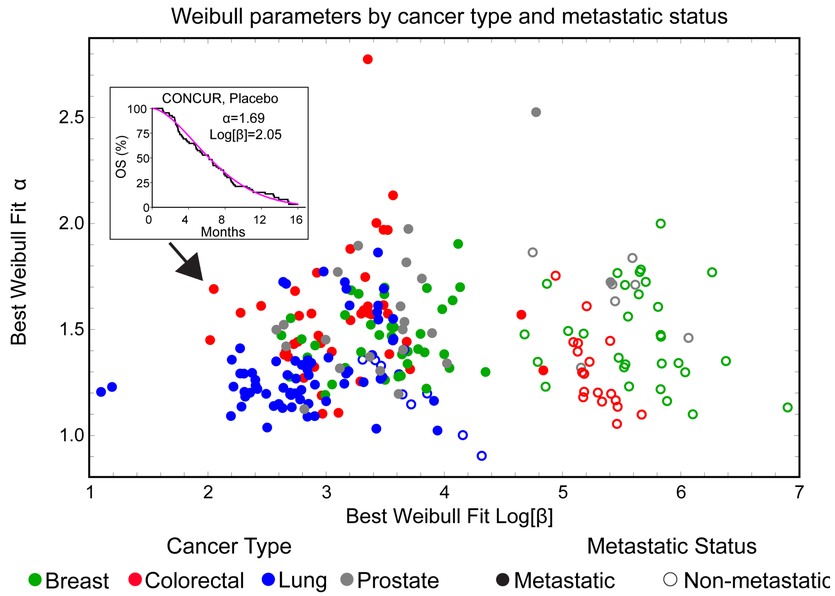


3) The quality of distribution fit was quantified using an R^2 value.

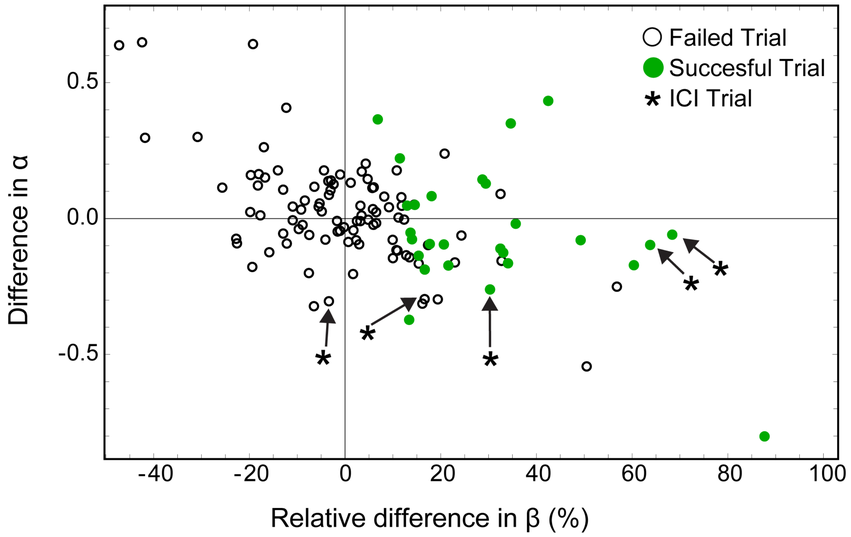


Results

Two-parameter Weibull distributions accurately describe overall survival (OS; R^2 =0.981) and visualize survival differences across cancer types.

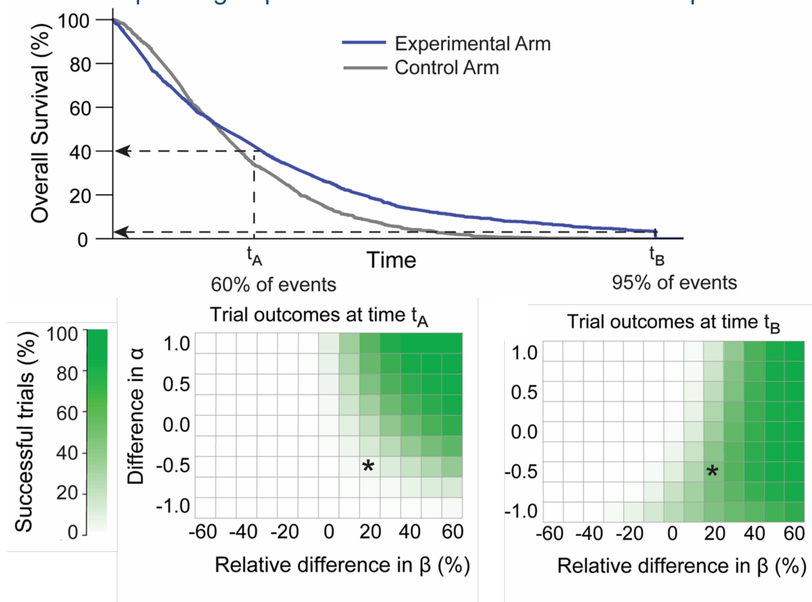


Trial success is assessed by using a statistical test that assumes that survival curves for treatment arms have the same shape (proportional hazards). We find that most trial arms have differences in α, the Weibull shape parameter, indicating a violation of this assumption.

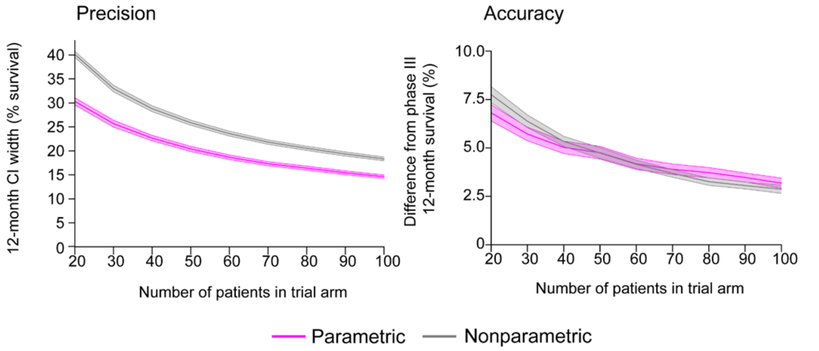


Simulation shows that for trials in which proportional hazards is violated, trial length impacts the likelihood of success.

*Plot corresponding to parameter values denoted in heatmaps.



Weibull (parametric) fitting of simulated data generates 12-month survival estimates with improved precision and equivalent accuracy as compared to conventional, non-parametric methods.



Conclusions & Data Availability

- Cancer patient survival is accurately described by two-parameter Weibull distributions.
- The duration of a trial has an underappreciated impact on its outcome.
- Use of parametric statistics could improve precision in early phase signal-seeking studies with small sample sizes.
- Data are available at **cancertrials.io.**
- Preprint and associated code are available at **bioRxiv 2021.05.14.442837**

Acknowledgements

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